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READING NOTE

RAPPORT

WaveNet : A Generative Model for Raw Audio

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1 Introduction

The paper *WaveNet : A Generative Model for Raw Audio*, co-written by researchers at Google DeepMind and Google, introduces WaveNet, a deep generative model designed to work directly on raw audio waveforms. The main goal was to explore high-resolution temporal audio generation techniques, inspired by recent advances in autoregressive neural generative models that have been used successfully to model complex distributions in images and text. The idea is to generate audio signals at a typical resolution of 16,000 samples per second or higher. Its arrival marked a real turning point in text-to-speech (TTS) synthesis by offering a very different approach from traditional methods based on predefined signal processing.

2 Why this paper ?

I chose this paper because I really wanted to get a deep understanding of one of the major innovation in high-resolution temporal audio generation. WaveNet’s results redefined the state of the art in TTS and opened up new research paths in audio generation. I wanted to see how I could combine my knowledge of deep learning and signal processing to grasp these innovations. Studying this paper let me see, at a fundamental level (individual sample), how neural networks can model complex temporal signals and the challenges that go with it (long-term dependencies, inference efficiency). It’s a solid foundation for anyone interested in deep learning for audio processing. Plus, WaveNet’s ability to model not just speech but also music, and even adapt to speech recognition, shows how flexible it is and its potential as a base model for many audio applications.

3 How does WaveNet work ?

3.1 Model architecture

WaveNet is a causal dilated convolutional network inspired by PixelCNN (used to generate images pixel by pixel), adapted here for audio. Formally, the model aims to approximate the joint distribution of an audio signal $(x_1, x_2, \dots, x_T)$ by factoring it into an autoregressive product of conditional distributions :

$$p(x_1, \dots, x_T) = \prod_{t=1}^T p(x_t \mid x_1, \dots, x_{t-1})$$

In other words, each sample x_t is generated based on all previous samples. To prevent any “future leakage” during training, the network uses causal convolutions : the output at time t depends only on inputs up to t , not beyond. Practically, this means shifting convolutional filters or using masks to block future information, which is crucial for audio since sound flows in time and “hearing” the future would make predictions unrealistic.

3.2 Causal dilated convolutions and receptive field

Modeling long-term temporal dependencies in audio needs a large input context. To do this without blowing up the parameter count, WaveNet uses dilated convolutions.

With a simple causal convolution, you'd need either many layers or very large filters to get a wide receptive field, making the model slow and heavy. Dilated convolution is like using a bigger filter derived by “spreading out” the original filter coefficients with zeros, so you get a larger window without increasing filter size. This lets the network “see” far-away information in time with only a few layers, keeping computation costs reasonable.

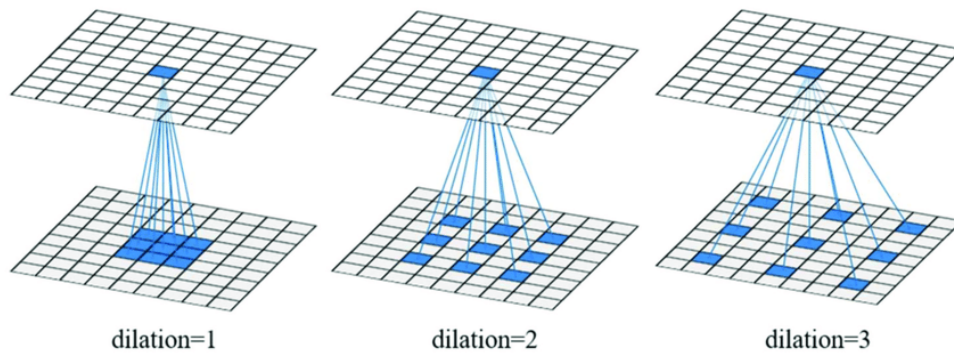


FIGURE 1 – Comparison between Standard Convolution and Dilated Convolution

By doubling the dilation rate at each layer, the receptive field grows exponentially with depth. For example, a stack of dilations 1, 2, 4, ..., 512 gives a receptive field of about 1024 samples per block. Stacking several of these blocks lets WaveNet cover a few hundred milliseconds of context, enough to capture local speech patterns.

3.3 Gated activation units

WaveNet borrows the gated activation mechanism from PixelCNN. Instead of a single nonlinearity (like ReLU) after convolution, each layer produces two parallel filters : one goes through a tanh, the other through a sigmoid σ . The element-wise product of these gives the gated output z . The sigmoid acts like a gate for the tanh, deciding what passes through, which models more complex interactions :

$$z = \tanh(W_{f,k} * x) \odot \sigma(W_{g,k} * x)$$

where $*$ is convolution, $\odot$ is element-wise multiplication, σ the sigmoid, and $W_{f,k}, W_{g,k}$ are learned filters for layer k .

3.4 Residual and skip connections

To ease training of such a deep network, WaveNet adds residual and skip connections at each layer. The layer's output z is added to its input (residual connection), so the layer only needs to learn the difference from the identity, which stabilizes training. Simultaneously, a transformed version of z (via a 1×1 convolution) is added to a cumulative sum of skip connections from all layers. At the end, this sum goes through two more transformations

(ReLU + 1×1 convolution) to produce the logits for a final softmax that gives the probability distribution over the next audio value. These skip connections ensure that contributions from far-away layers reach the output intact, improving gradient flow and modeling quality. As illustrated in this figure :

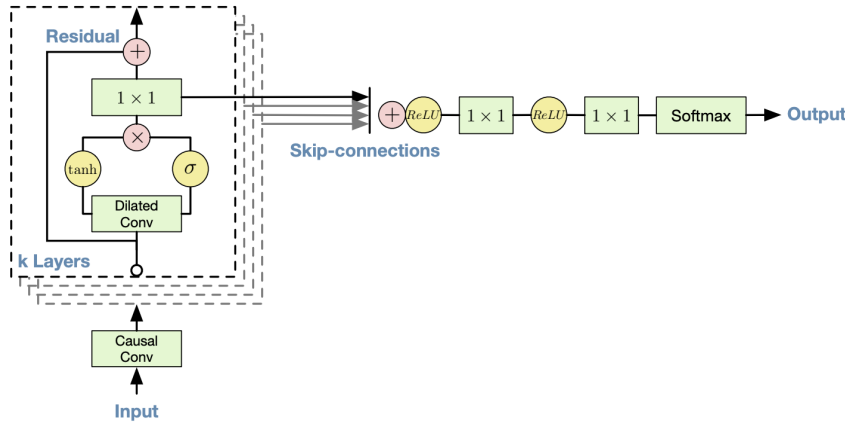


FIGURE 2 – Residual and skip connections (Source : the paper)

3.5 Quantization and output

WaveNet's outputs are 8-bit quantized audio samples (256 possible values) instead of 16-bit (65,536 values). The authors choose this to keep the softmax size manageable while preserving audio fidelity via μ -law companding :

$$f(x_t) = \text{sign}(x_t) \frac{\ln(1 + \mu|x_t|)}{\ln(1 + \mu)}, \quad -1 < x_t < 1, \mu = 255.$$

WaveNet converts the continuous signal into integer indices from 0 to 255 (via μ -law encoding) as input, and predicts at each step a 256-way softmax for the next sample. During inference, you actually sample from this distribution, then apply the inverse μ -law to get the audio value. This probabilistic process gives varied intonations and a more natural long-term result compared to a deterministic output.

3.6 Training vs. generation

WaveNet is trained by standard supervised learning, maximizing the likelihood of the training data (real audio). Since the factorized form gives an explicit probability, you can use cross-entropy loss between the predicted distribution and the real next sample, which is equivalent to maximizing log-likelihood. Training is parallelized over known audio sequences : at each time t , the model is given the true previous value x_{t-1} (teacher forcing) so all predictions in a sequence can be computed at once. In generation, however, the model has no future targets : it must produce one sample at a time, feeding its own previous prediction back to generate the next. This autoregressive process is computationally expensive (especially at 16 kHz !), so WaveNet's audio synthesis is relatively slow. But this

step-by-step generation is what allows the model to create coherent, realistic waveforms, with each new sample reflecting subtle past variations.

3.7 Conditioning mechanisms

WaveNet can be trained unconditionally (generating audio from scratch) or conditioned on extra inputs.

3.7.1 Global conditioning

A single conditioning vector is used for the whole waveform (e.g., speaker identity or language) :

$$z = \tanh(W_{f,k} * x + V_{f,k}^T h) \odot \sigma(W_{g,k} * x + V_{g,k}^T h)$$

where h is the global conditioning vector and $V_{f,k}, V_{g,k}$ are learned projections.

This vector is added as a bias at each layer, letting the model switch speaker or style.

3.7.2 Local conditioning

A time-series of features is provided alongside the audio. These features are :

- Up-sampled (via transposed convolutions) to match the audio sample rate,
- Injected into every convolutional layer through learned 1×1 convolutions, added to the gated activations.

$$z = \tanh(W_{f,k} * x + V_{f,k} * y) \odot \sigma(W_{g,k} * x + V_{g,k} * y),$$

where y is the up-sampled feature sequence.

In TTS, this local conditioning lets WaveNet act as a neural vocoder, turning a coarse intermediate representation into high-fidelity audio. The paper shows that using linguistic features for conditioning greatly improves voice naturalness compared to traditional parametric or concatenative methods.

3.8 In summary, WaveNet combines...

1. An autoregressive model : each sample is generated based on previous ones.
2. Causal dilated convolutions : efficiently modeling long-term dependencies.
3. Flexible conditioning mechanisms : guiding the generation process.

This makes it especially well suited for generating very long raw audio sequences that need both short- and long-term structure.

4 Critical analysis of methodology and paper impact

WaveNet was evaluated on tasks like multi-speaker TTS, text-conditioned TTS, music generation, and even speech recognition. The model produces exceptionally natural speech, outperforms classical approaches (parametric, concatenative, RNNs), learns multiple speaker characteristics, generates harmonic music, and can be adapted for discriminative tasks.

Methodological strengths

- **Exceptional naturalness in TTS** : with a record MOS score (>4.0) and over 50% reduction in gap to human speech.
- **End-to-end approach** : operates directly on raw waveform, without a vocoder or intermediate feature extraction, eliminating common artefact sources.
- **High flexibility** : handles both speech and music, supports global (speaker) and local (linguistic/F0) conditioning, and adapts to discriminative tasks.
- **Efficient dilated convolutions** : capture long-term dependencies without huge computational cost.
- **Optimizable pipeline with computable log-likelihood** : allowing rigorous hyperparameter tuning.

Limitations

- **Costly sequential inference** : unlike parallelizable training, audio generation is slow since each sample is generated one by one, limiting real-time use.
- **Limited long-term memory** : despite large receptive fields, WaveNet struggles to maintain coherence over very long scales (full songs, long sentences).
- **Heavy dependence on receptive field size** : quality hinges on this choice, which can get expensive for complex tasks.
- **Prosody needs better conditioning** : receptive field alone can't capture all intonation subtleties without F0 or similar features.

Impact and legacy

WaveNet marked a major shift in audio synthesis and generation. It laid the groundwork for modern end-to-end approaches, proved you could generate high-quality raw audio, and inspired many follow-up works aimed at speeding up inference (Parallel WaveNet, WaveRNN. . .) or integrating neural audio generation into full TTS systems like Tacotron. Its methodology has had huge influence in both industry and research.

In short, WaveNet redefined audio synthesis standards by delivering a major leap in naturalness at the cost of high inference complexity. Its flexibility and universality made it a go-to reference, widely adopted and extended in the community.